

Asymmetric Capital Control: Short Summary

Codes in parentheses refer to results in the proofs document, which carries an index of them.

Two banks facing identical regulation can run visibly different capital policies. A bank that treats a shortfall as worse than an equivalent **surplus** is good will hold more capital, pay out later, and recapitalise sooner than the rules alone require—but so will a bank simply responding to tight regulation. The two are not distinguishable by inspection, yet they predict different behaviour when the rules change.

This paper solves a model in which they can be told apart, and the natural place to look is the wrong one. A bank manages its capital **buffer** under a **regulatory cap**, paying out at a **dividend barrier** and raising capital in lump sums at a **recapitalisation trigger**, with shortfalls below a **reference level** penalised by one asymmetry parameter. At its neutral value the problem exactly nests the **risk-neutral benchmark** (NST), so the parameter captures asymmetry beyond what regulation and costs alone produce.

The **definitional evaluation** of the **deficit-to-surplus** volatility ratio—the second-moment measure one would instinctively reach for—is a constant of **cap geometry**, carrying no information about the preference whatsoever (SCG). The thresholds carry it, and both responses are obtained in closed form: a more shortfall-averse bank holds a larger **buffer** before paying out and recapitalises earlier (TCS). The **fixed issuance cost** moves the **recapitalisation trigger** the opposite way (KCS), and that opposition is decisive — it makes the Jacobian of the two thresholds in the two parameters non-singular, so preference and cost are jointly identified from where the bank commits to act (IDN). The identification turns on a sign, not on a comparison of magnitudes.

Preference is legible in the boundaries a bank commits to, not in the volatility its behaviour produces. To our knowledge no firm-level model combines singular payout control with impulse **recapitalisation** under an asymmetric preference; the recent preference-generalised dividend literature remains singular-only (Chen et al., 2026; Cao et al., 2026).

References

- Jingyi Cao, Dongchen Li, Virginia R. Young, and Bin Zou. Singular dividend control under mean-variance preferences, 2026. arXiv:2511.08433; accepted, SIAM Journal on Control and Optimization. Volume and page numbers not yet verified against the journal.
- An Chen, Seyoung Park, and Hoi Ying Wong. Robust dividend policy: Equivalence of Epstein-Zin and Maenhout preferences, 2026. arXiv:2406.12305.